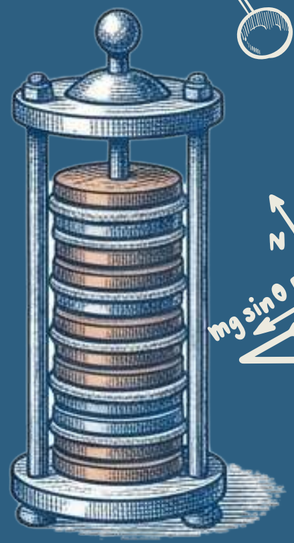
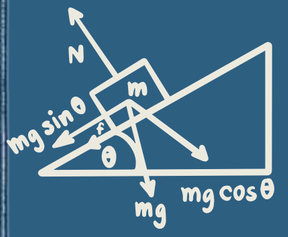
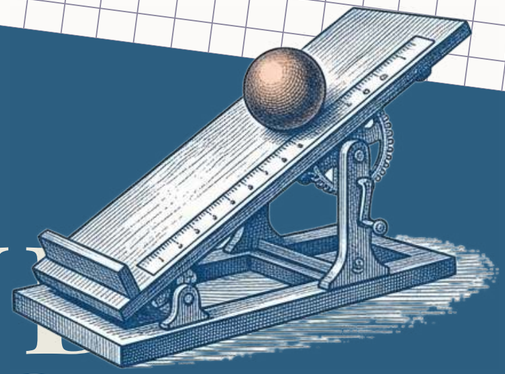
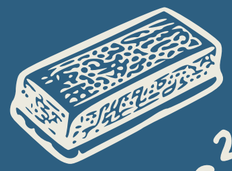
SCIENCE, TECHNOLOGY, AND NATION-BUILDING







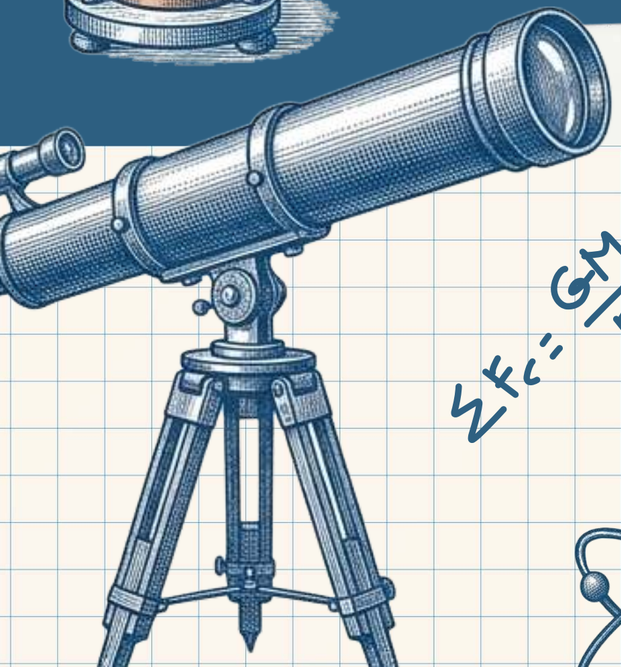


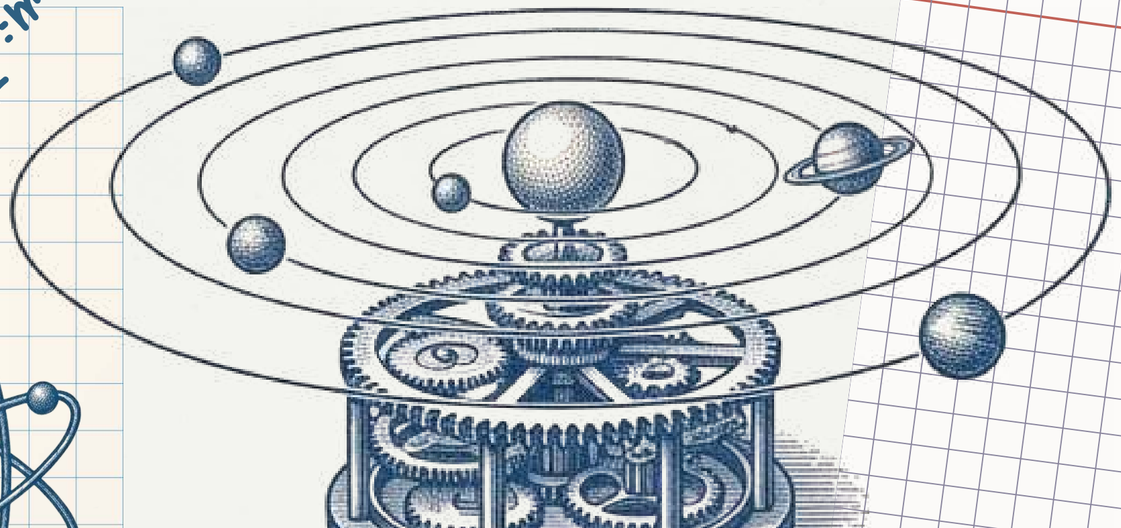


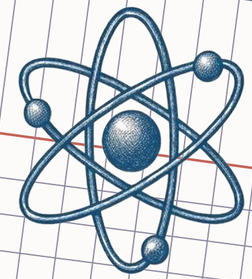
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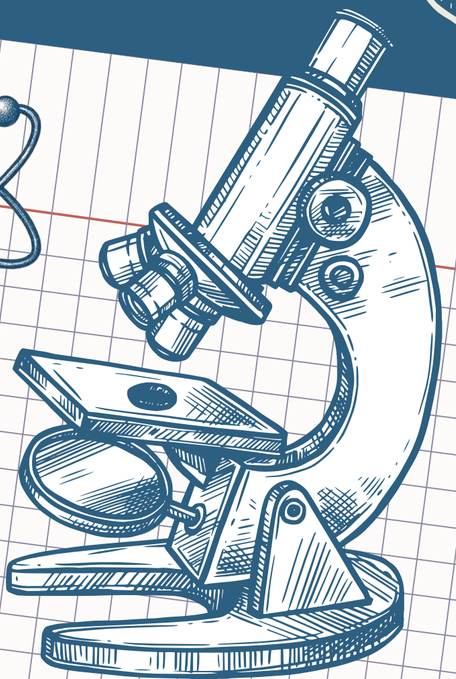
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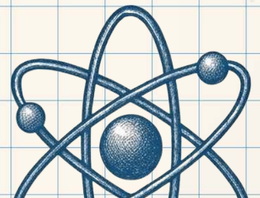
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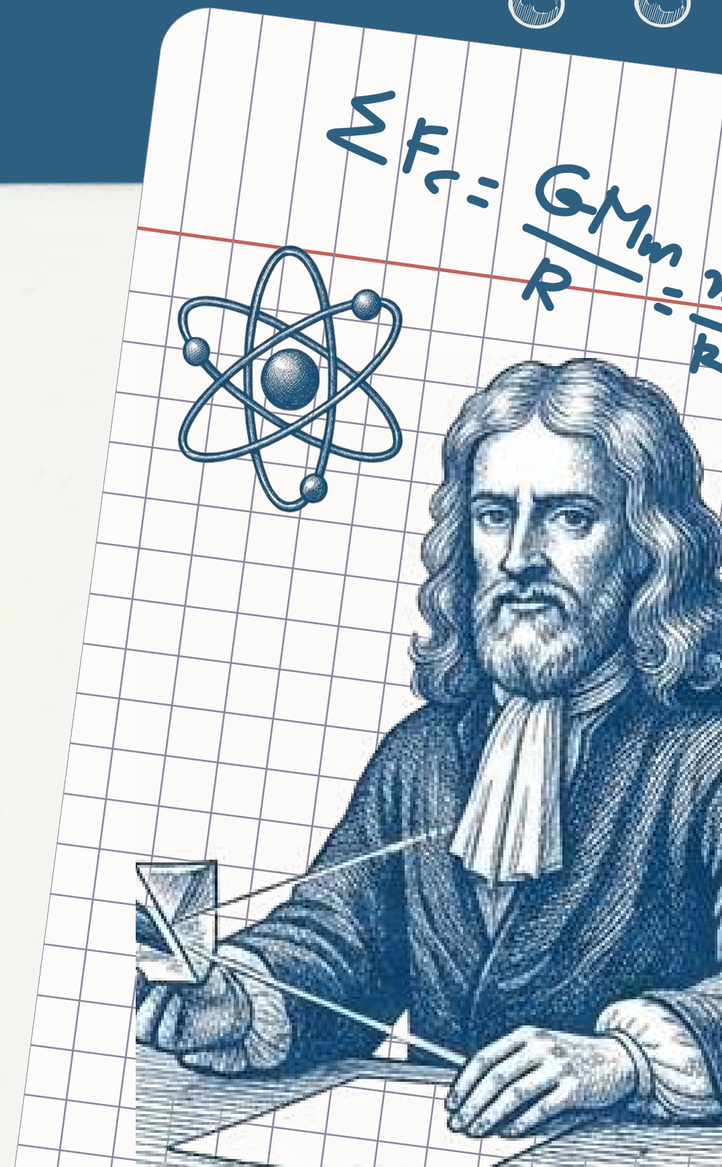






WHAT IS SCIENCE & TECHNOLOGY?





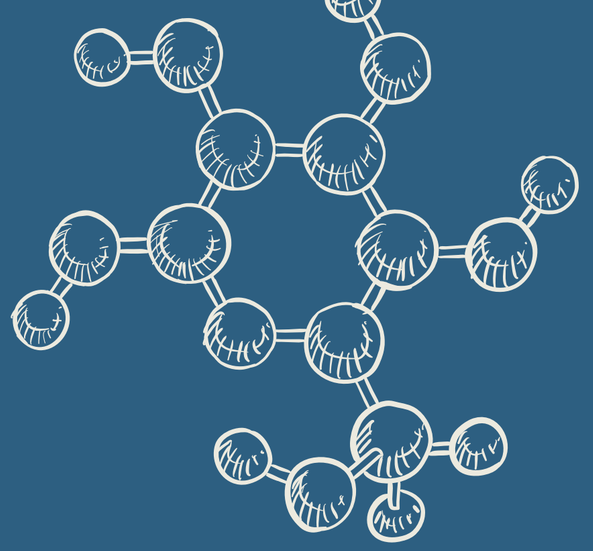
SCIENCE

- Systematic study of the physical world through observation and experimentation.
- Focuses on understanding concepts, principles, and natural laws.

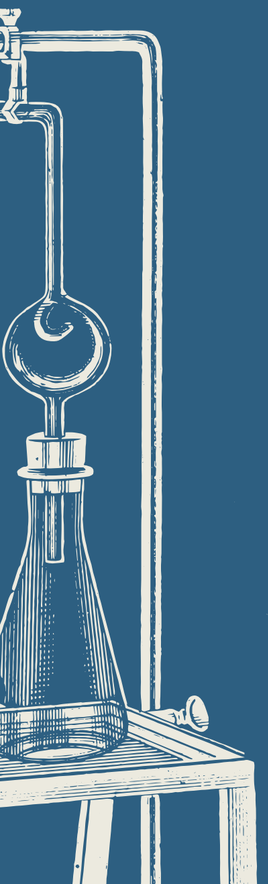
TECHNOLOGY

- Practical application of scientific knowledge.
- Includes tools, systems, and methods to solve real-world problems and improve quality of life.

WHAT IS NATION-BUILDING?



It is constructing or structuring a national identity through the power of the state. It aims to unify people of diverse culture, language, and religion under the same state so that it remains politically stable and viable in the long run.





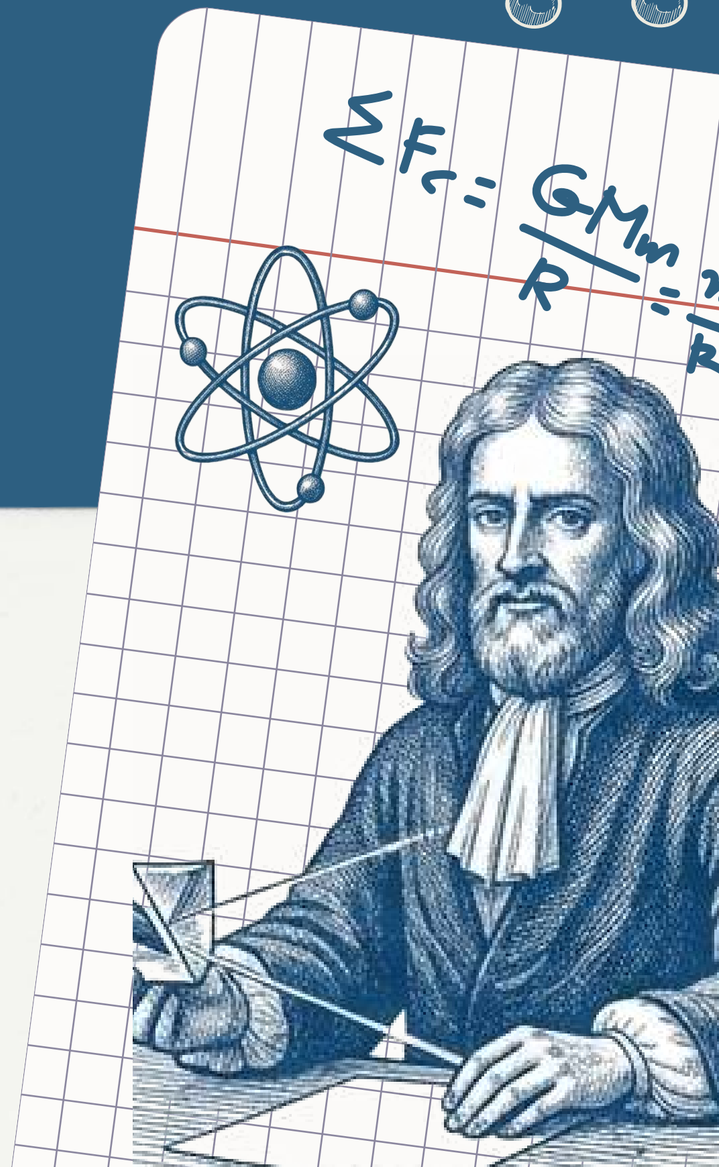
PHILIPPINE NATION-BUILDING

According to Morte (2022), Philippine nation-building relies on three main pillars:

1. **Political Integration:** Having fair laws and good governance.
2. **Economic Development:** Creating opportunities for sustainable growth.
3. **Social Cohesion:** Active citizen participation and inclusive policies that empower every Filipino.

HISTORICAL BACKGROUND

OF SCIENCE AND TECHNOLOGY IN THE PHILIPPINES







PRE-COLONIAL PERIOD (BEFORE 1565)

- Contrary to popular belief, early Filipinos were already skilled, intelligent, and civilized long before the arrival of the Spanish.

- Highly resourceful in agriculture, maritime exploration, metalwork, construction, and herbal medicine.





EARLY INDIGENOUS TECHNOLOGY

Early Filipinos utilized local natural resources—such as clay, bamboo, hardwood, and mineral ores—to develop specialized tools, goods, and infrastructure for everyday life and trade.

HERBAL MEDICINE

Healthcare practices utilizing the archipelago's rich biodiversity, led by traditional healers (*albularyo/herbolaryo*).



Malunggay
— Immunity, breast milk,
— Boil for tea



Lagundi
— Cough, asthma, fever
— Boil leaves for tea



Sambong
— Kidney stones, UTI
— Boil and drink 3x daily



Oregano
— Cough, sore throat
— Boil with honey



Yerba Buena
— Stomach ache, headache
— Crush or boil



Bayabas
— Wounds, mouth sores, LBM
— Boil for wash or gargle



Tawa-tawa
— Dengue, low platelets
— Boil whole plant



Tsaang Gubat
— Stomach ache, diarrhea
— Boil for tea



Tanglad
— Fever, colds, calming
— Boil stalks for tea



Sabila
— Burns, wounds, hair, skin
— Apply gel directly

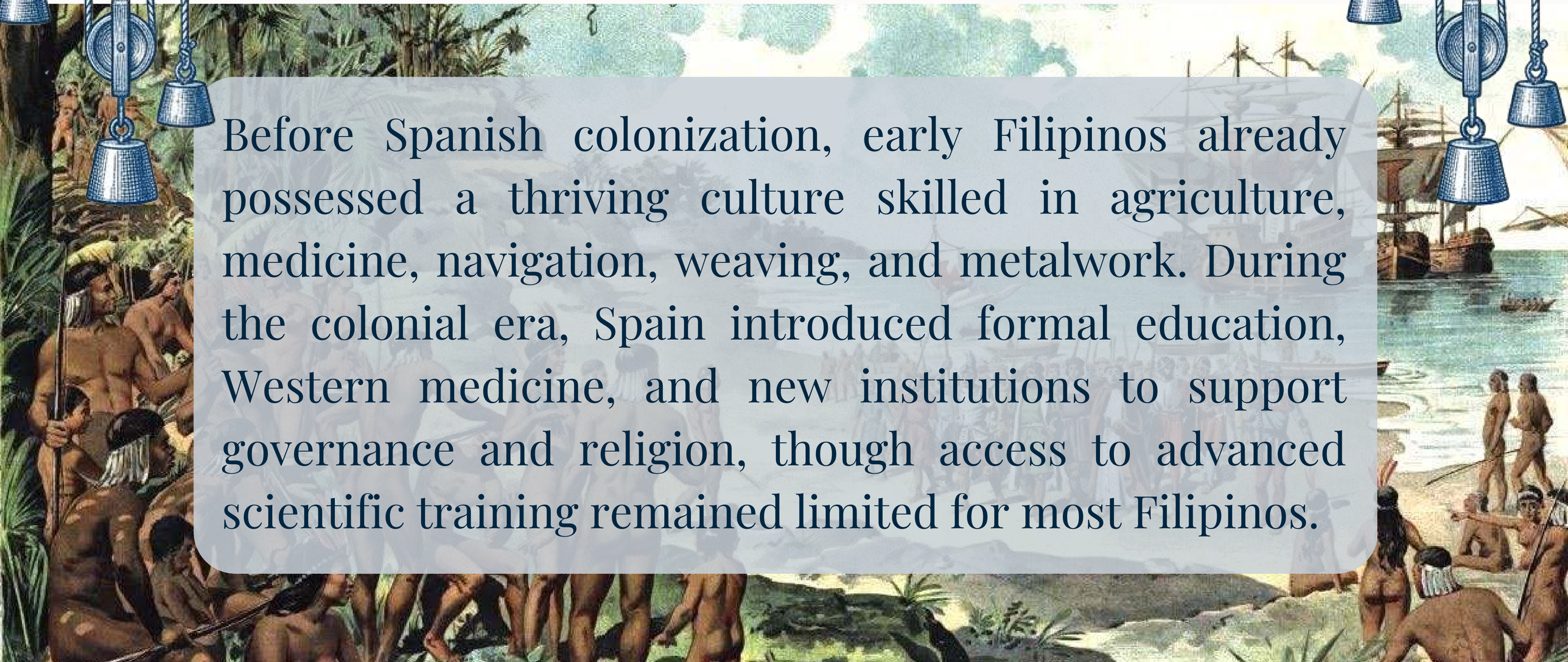


AGRICULTURAL PRACTICES

Pre-colonial farming in the Philippines was not just about planting. It was also a way of bringing life to the community. Filipino farmers would clear the weeds and plow the land to make the soil fertile. With that, they were able to plant things like rice, ube, and taro.

- **Vegetation Clearing** - it prepares previously overgrown or forested land for crop cultivation, improves soil conditions and enhances irrigation systems.
- **Soil tilling** - Improve the quality of soil for better crop yield
- **Irrigation System** - maintaining stable water supply for crops. They create canals and terraces to make water flow directly in rice fields to keep it.

SPANISH COLONIAL ERA (1565 - 1898)



Before Spanish colonization, early Filipinos already possessed a thriving culture skilled in agriculture, medicine, navigation, weaving, and metalwork. During the colonial era, Spain introduced formal education, Western medicine, and new institutions to support governance and religion, though access to advanced scientific training remained limited for most Filipinos.

INTRODUCTION OF FORMAL SCIENCE EDUCATION AND MODERN MEDICINE

Beginning in 1565, Spain introduced a church-linked, European-style education system designed primarily to spread Christianity and Spanish culture in the Philippines.

ROLES OF SPANISH MISSIONARIES AND RELIGIOUS ORDERS

Spanish religious orders (Augustinians, Franciscans, Jesuits, and Dominicans) established parishes and schools across the Philippines, using them as community hubs to spread education, Catholicism, and Spanish culture.



ESTABLISHMENT OF ACADEMIC INSTITUTION



University of Santo Tomas

The University of Santo Tomas (UST), founded in 1611, was originally located in Intramuros, Manila. It was first called Colegio de Nuestra Señora del Santísimo Rosario and later renamed Colegio de Santo Tomas. It became a major center of higher education during the Spanish colonial period.

ESTABLISHMENT OF INFRASTRUCTURE

- **Fort Santiago, Manila –** built in 1571, part of the walled city's defense structure.
- **San Agustin Church, Intramuros –** Stone church in intramuros, serving religous and civic life across centuries.
- **Puente de España, Manila –** completed in 1630 and reconstructed in 1875, known as Bridge of Spain connecting transportation and communities.
- **Port of Manila –** a maritime center supporting the movement of trade in goods between the Philippines and other countries.













AMERICAN COLONIAL ERA & COMMONWEALTH PERIOD:

AMERICAN COLONIAL ERA & COMMONWEALTH PERIOD:

Education and Training

- **Thomasites (1901):** Establish modern public schools focus on science, agriculture, & English.
- **UP Founded (1908):** Strengthened higher educati and research.
- **Scholarships:** More Filipino studied abroad in engineerin and medicine.

Public Health and Medicine

- Improve sanitation systems, clean water supply, and widespread vaccination programs were introduced.
- Quarantine stations and health boards were established to contro outbreaks like cholera and smallpox.

Commonwealth Period (1935-1946)

- **1935 Constitution:** State mandated to promote scientific research and invention.
- **NRCP (1933):** Created to coordinate national research.
- **Interruption:** Progress halted during WWII and Japanese occupation.

POST-WORLD WAR II & CONTEMPO RARY ERA:

After gaining independence in 1946, the Philippines rebuilt its scientific infrastructure and gradually established national policies to guide science and technology development.

Post-War Recovery (1946-1950s)

- The Bureau of Science was recognized into the Institute of Science in 1946 to lead post-war scientific recovery.

- Science Act of 1958 (RA 2067) - a landmark law that created the National Science and Development Board (NSDB) the central planning agency for S&T.

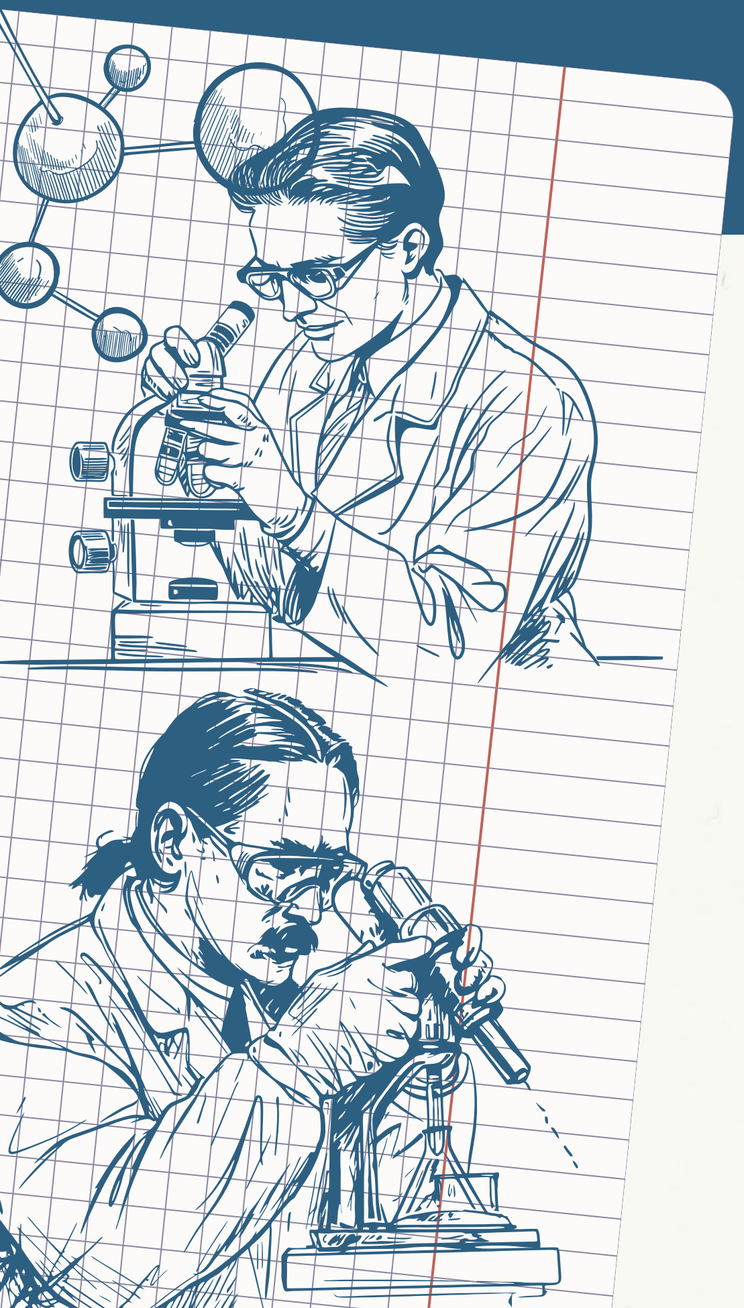
POST-WORLD WAR II & CONTEMPORARY ERA:

Institutional Growth (1960s-1980s)

From the 1960s to the 1980s, specialized research institutes were established under the NSDB to focus on agriculture, textiles, metals, forest products, and nuclear science. The board was reorganized as the National Science and Technology Authority (NSTA) in 1982 to streamline S&T coordination, eventually becoming the cabinet-level Department of Science and Technology (DOST) in 1987. By 1988, the government launched its first Science and Technology Master Plan to drive industrial development and enhance quality of life.

Contemporary Era (1990s-Present)

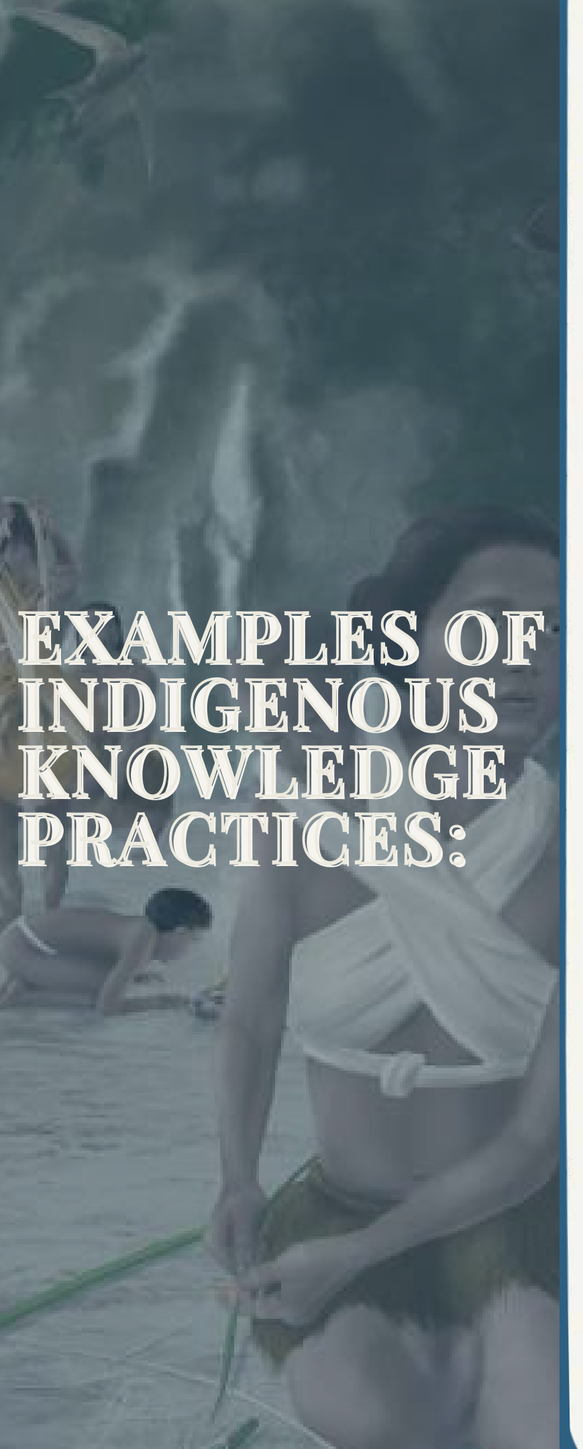
In the contemporary era, key legislation—such as the Balik Scientist Act, the Philippine Innovation Act, and the Innovation Startup Act—was enacted to strengthen national innovation. Modern research shifted toward disaster preparedness, climate change adaptation, renewable energy, and space technology. Today, specialized DOST agencies like PHIVOLCS and PAGASA deliver critical early warning systems for natural hazards, while long-term initiatives like PAGTANAW 2050 pave the way for a technology-driven Philippines.



INDIGENOUS SCIENCE

It refers to the traditional knowledge systems, practices, and community-based technologies developed by Indigenous peoples through long-term observation of and interaction with their environment.

In the Philippines, Indigenous Knowledge Systems and Practices (IKSPs) help communities manage forests, farms, water, biodiversity, disasters, and climate-related risks



EXAMPLES OF INDIGENOUS KNOWLEDGE PRACTICES:

1. Traditional agriculture
2. Weather forecasting
3. Herbal medicine
4. Crop rotation and intercropping
5. Forest and water management



Importance to Philippine Science and Technology

- Provides locally tested knowledge about agriculture, health, disaster preparedness, and environmental conservation. Modern science can strengthen these practices through research, technology, education, and policy support.
- Philippine government recognizes the value of integrating Indigenous knowledge into climate adaptation and national planning.
- The country's National Adaptation Plan highlights Indigenous knowledge alongside science-based approaches to create inclusive, culturally appropriate, and effective solutions to climate change

GOVERNMENT POLICIES AND PROGRAMS ADVANCING

Science and Technology in the Philippines





BALIK SCIENTIST PROGRAM:

The Balik Scientist Program (BSP) is a government initiative led by the Department of Science and Technology (DOST) that encourages Filipino scientists, researchers, and technology experts working abroad to share their knowledge and expertise with Philippine institutions.

GOVT POLICIES & PROGRAMS IN PH

OBJECTIVES:

1. Address *brain drain*
2. Strengthen research and development
3. Develop Filipino scientists and researchers
4. Support national research priorities

BENEFITS TO SCIENTIST

SHORT TERM • Minimum duration of 15 days to a maximum duration months • To be served within 1 year from the date of the award	• Daily allowance (USD 200/day) or the UNDP rate whichever is higher • Health and medical (including dental) insurance - Php 250,000 per illness • Accident/travel insurance - up to Php 3,000,000 coverage • GIA (Grants-In-Aid) funding for research (proposal driven) • Exemption from PRC requirements
MEDIUM TERM • Duration of more than 6 months but not to exceed 1 • To be served within 2 years from the date of the awa	• Round trip domestic airfare and accommodation for two (2) local conferences • The rest are same with short term
LONG TERM • Duration of more than one (1) year to three (3) years, su to renewal to be determined by DOST • To be served within five (5) years from the date of the a	• One (1) international round trip airfare for the Scientist, spouse and dependent/s • Special non-immigrant visa for the Scientist, spouse and dependent/s • Relocation allowance - Php 250,00 • Housing allowance - up to Php 40,000 per month • Salary equivalent to at least SG 24 - from Php 102,603 to Php 114,301 per month • Round trip domestic airfare and accommodation for three (3) local conferences • The rest are same with short term

BENEFITS TO THE PHILIPPINES

- Stronger Research Institutions
- Development of local talent
- Improved public welfare and services
- Economic growth and innovation

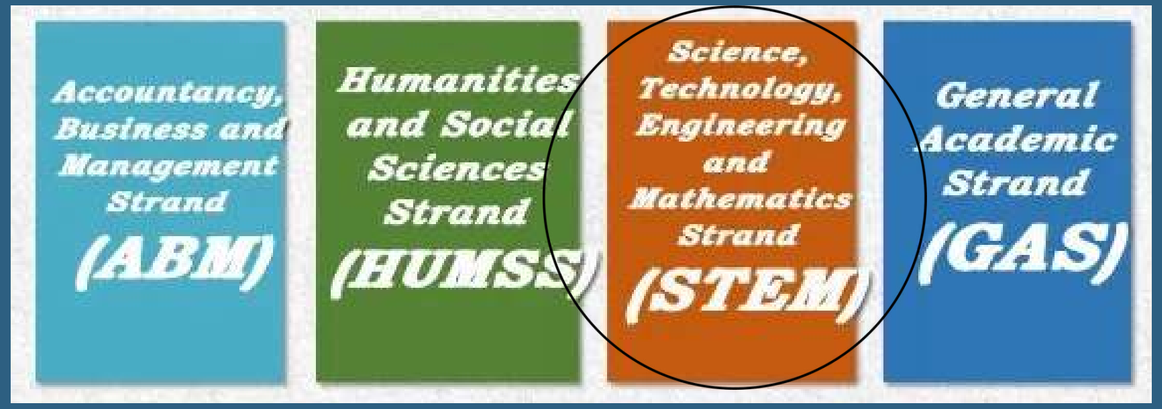
IMPACT:

- The BSP has contributed to the advancement of Philippine science by bringing international expertise into local research institutions, strengthening scientific capabilities, and promoting innovation across different sectors. Since the establishment in 1975 up to 2025.
- The program has welcomed 697 returned Filipino scientists, with a total of 933 engagements recorded. These scientists have contributed to research and innovation in health, food safety, biotechnology, renewable energy, agriculture, and coastal resilience.
- Through research collaboration, mentorship, and technology transfer, the program helps develop Filipino researchers, strengthen local institutions, and generate science-based solutions to national challenges. Overall, it helps build a stronger Philippine research community and encourages the use of scientific knowledge to improve public health, environmental protection, and national development.

K TO 12 STEM PROGRAM:

Science, Technology, Engineering, and Mathematics (STEM) strand is an academic strand under the Senior High School program of the Philippine K-12 curriculum. Focusing on advanced science and mathematics subjects to develop students' analytical thinking, problem solving and research skills. It prepares learners for higher education and careers in fields related to science, technology, and engineering.

ACADEMIC Track (4 Strands)



OBJECTIVES:

1. Develop scientific and mathematical competencies
2. Prepare students for higher education and STEM related career
3. Promote research, innovation, and analytical thinking

IMPACT:

- The K-12 STEM program contributes to the country's science and national development by preparing students for STEM related college programs/courses and careers, while developing skills in research, experimentation, and problem-solving.
- Part of the STEM curriculum is the Research/Capstone project which trains students to collect and analyze data and present scientific findings, while DepEd and DOST's partnership support teacher training, curriculum development, and student engagement in science and technology.
- Help build a future workforce capable of contributing to Philippine research and innovation.

OTHER KEY S&T INITIATIVES:

Research grant programs

A. Grants-in-Aid (GIA) program (National Research Council of the Philippines)

Funds original and problem-oriented research in science and humanities that addresses national priorities.

Eligibility/Requirements:

- NRCP regular or associate membership
- Research proposal aligned with NRCP/DOST priority areas
- Application form, researcher's CV, and institutional endorsement
- No outstanding technical or financial obligations to NRCP/DOST

OTHER KEY S&T INITIATIVES:

B. DOST Grants-in-Aid for Sectoral Research (DOST)

Funds research projects in specific fields:

- PCAARRD (Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development) - agriculture, aquatic resources, forestry, and natural resources
- PCHRD (Philippine Council for Health Research and Development) - health and biomedical research
- PCIEERD (Philippine Council for Industry, Energy, and Emerging Technology Research and Development) - Industry, energy, and emerging tech

Eligibility/Requirements:

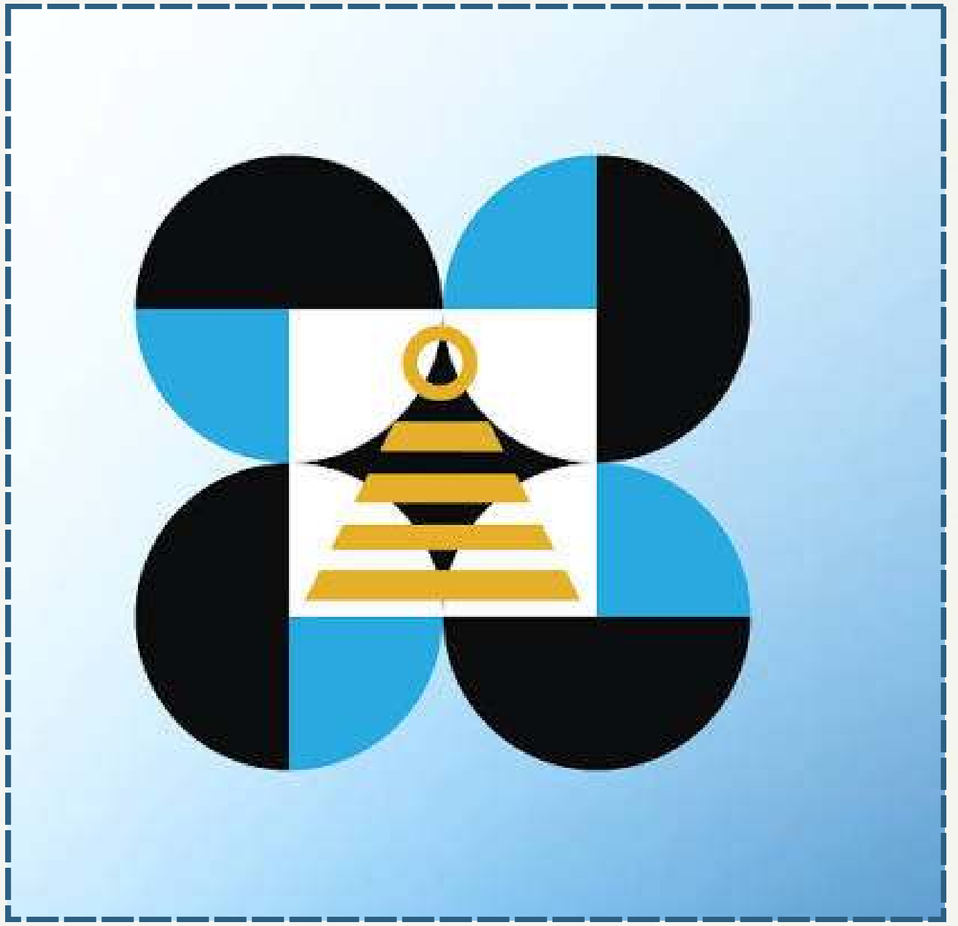
- Proposals are to be submitted through eligible universities, government research institutions, or qualified private-sector applicants
- Project must match the council's research priorities

DOST SCHOLARSHIP PROGRAMS

A. DOST-SEI Merit Scholarship
Financial support for academically talented Filipino students pursuing priority science and technology courses/programs

Eligibility/Requirements:

- Natural born Filipino citizen
- Graduating Grade 12 student in the STEM strand or top 5% of a non-STEM class
- Good moral character and good health
- Must pass the scholarship selection process and enroll in an eligible priority degree program



DOST SCHOLARSHIP PROGRAMS



B. Republic Act No. 7687 "Science and Technology Scholarship Act of 1994"
A scholarship for talented students from economically disadvantaged families pursuing priority STEM related degree program

Eligibility/Requirements:

- Natural born Filipino citizen
- Meets the program's family socioeconomic criteria
- Graduating Grade 12 student in the STEM strand or top 5% of a non-STEM class
- Passes the selection process

TECHNOLOGY COMMERCIALIZATION EFFORTS

A. Technology Transfer and Entrepreneurship Collaboration and Harmonization of Growing Regional Opportunities on Wealth Creation (TECHGROW) Initiative that connects researchers, universities, government agencies, and businesses to help turn research outputs into marketable products and services

B. Small Enterprise Technology Upgrading Program (SETUP)
Provides technology assistance, equipment support, technical consultancy, training, and product development services to help micro, small, and medium enterprises improve their operations


ICE
BREAKER




BRING
ME...



COMPARISONS BETWEEN THE TWO PHILIPPINE DEVELOPMENT PLANS

PHILIPINE DEVELOPMENT PLAN (2017-2022)

The Philippine Development Plan of 2017-2022 Goals:

1. Build a prosperous, predominantly middle-class society where no one is poor.
2. Promote a long and healthy life.
3. Become smarter and more innovative.
4. Build a high-trust society.

These goals were based from collective long-term vision of Filipinos to have: “matatag, maginhawa, at panatag na buhay”.

Key Areas from Philippine Development Plan (2017-2022):

1. Enhancing the social fabric (malasakit)
2. Inequality-reducing transformation (pagbabago)
3. Increasing growth potential (patuloy na pag-unlad)
4. Enabling and supportive economic environment
5. Foundations for sustainable development

KEY AREAS FROM PHILIPPINE DEVELOPMENT PLAN (2017-2022):

1. **Enhancing the social fabric (malasakit)**
   - Greater trust in public institutions and society
   - People-centered, clean, and efficient government
   - Justice is swift and fair
   - Awareness about and respect for cultural diversity
2. **Inequality-reducing transformation (pagbabago)**
   - Greater trust in public institutions and society
   - People-centered, clean, and efficient government
   - Justice is swift and fair
   - Awareness about and respect for cultural diversity
3. **Increasing growth potential (patuloy na pag-unlad)**
   - Adoption of modern technology
   - Innovation is encouraged
   - Harmonized research and development agenda
   - Interventions to manage population growth
   - Investments for human capital development



KEY AREAS FROM PHILIPPINE DEVELOPMENT PLAN (2017-2022):

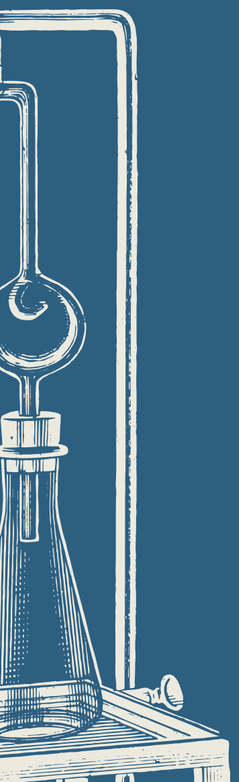
4. **Enabling and supportive economic environment**

- Adoption of modern technology
- Innovation is encouraged
- Harmonized research and development agenda
- Interventions to manage population growth
- Investments for human capital development

5. **Foundations for sustainable development**

- Attaining just and lasting peace
- Ensuring security, public order, and safety
- Accelerating infrastructure development
- Ensuring ecological integrity, clean and healthy environment.

These four pillars represent the essential baseline conditions required for a society to progress without collapsing.





PHILIPPINE DEVELOPMENT PLAN GOALS (2023 - 2028)

ECONOMIC AND SOCIAL TRANSFORMATION

FOR A PROSPEROUS, INCLUSIVE, AND RESILIENT SOCIETY



PHILIPPINE DEVELOPMENT PLAN (2023 - 2028)

The Philippine Development Plan (PDP) 2023–2028 focuses on transforming the Philippine economy and society to achieve inclusive and sustainable development. It aims to create more quality jobs, improve people's living conditions, and strengthen the country's ability to respond to economic and environmental challenges.

Social Transformation: Develop and Protect Capabilities of Individuals and Families

Economic Transformation: Transform Production Sectors to Generate More Quality Jobs and Competitive Products



PHILIPPINE DEVELOPMENT PLAN GOALS (2023 - 2028)

The goal of the PDP 2023 2028 socioeconomic goals:

1. Maintain an annual economic growth rate between 6.0 to 7.0 percent in 2023 and between 6.5 to 8.0 percent from 2024 to 2028.
2. Create more, better, and more resilient jobs.
3. Keep food and overall prices low and stable.
4. Enforce fiscal discipline.
5. Transform the production sectors through innovation.
6. Reduce poverty incidence from 18 percent in 2021 to between 8 to 9 percent by 2028

These goals were also based on AmBisyon Natin 2040, which aims to achieve a "matatag, maginhawa, at panatag na buhay."

PHILIPPINE DEVELOPMENT PLAN GOALS (2023 - 2028)

STRATEGIES AND APPROACHES FOR ECONOMIC TRANSFORMATION

Economic transformation focuses on improving agriculture, industry, and services to create more quality jobs and competitive products. The government encourages businesses to adopt modern technology, strengthen productivity, and develop innovative products. These strategies aim to increase economic opportunities and make Philippine industries more competitive in local and international markets.

1. **MODERNIZE AGRICULTURE AND AGRIBUSINESS**
2. **REVITALIZE INDUSTRY**
3. **REINVIGORATE SERVICES**

PHILIPPINE DEVELOPMENT PLAN GOALS (2023 - 2028)

STRATEGIES AND APPROACHES FOR ECONOMIC TRANSFORMATION

1 MODERNIZE AGRICULTURE AND AGRIBUSINESS

This strategy focuses on improving agricultural productivity by using modern technology and better farming methods. It also aims to increase the value of agricultural products and encourage farmers to produce different crops.

2 REVITALIZED INDUSTRY

It aims to strengthen local industries by encouraging innovation and the use of modern technology. It also promotes partnerships between businesses and academic institutions to develop better products and services. These strategies will help Philippine industries become more competitive in local and international markets.

3 REINVIGORATE SERVICES

Improving productivity and creating more job opportunities are the main goals of this strategy. The government will strengthen tourism, creative industries, information technology, and business services through partnerships and innovation.

PHILIPPINE DEVELOPMENT PLAN GOALS (2023 - 2028)

STRATEGIES AND APPROACHES FOR SOCIAL TRANSFORMATION

Social transformation focuses on improving the capabilities of individuals and families through better education, healthcare, and social protection. It also promotes employment opportunities and skills development to help Filipinos adapt to changes in the economy. These efforts aim to reduce poverty and ensure that more people benefit from economic growth.

1

PROMOTE HUMAN AND SOCIAL DEVELOPMENT

2

REDUCE VULNERABILITIES AND PROTECT PURCHASING POWER

3

INCREASE INCOME-EARNING ABILITY

PHILIPPINE DEVELOPMENT PLAN GOALS (2023 - 2028)

STRATEGIES AND APPROACHES FOR SOCIAL TRANSFORMATION

1 PROMOTE HUMAN AND SOCIAL DEVELOPMENT

This strategy focuses on improving agricultural productivity by using modern technology and better farming methods. It also aims to increase the value of agricultural products and encourage farmers to produce different crops.

2 REDUCE VULNERABILITIES AND PROTECT PURCHASING POWER

The government aims to protect Filipinos from economic difficulties by ensuring access to affordable, safe, and nutritious food. It also focuses on strengthening social protection programs, such as financial assistance, insurance, and other support for vulnerable individuals and families.

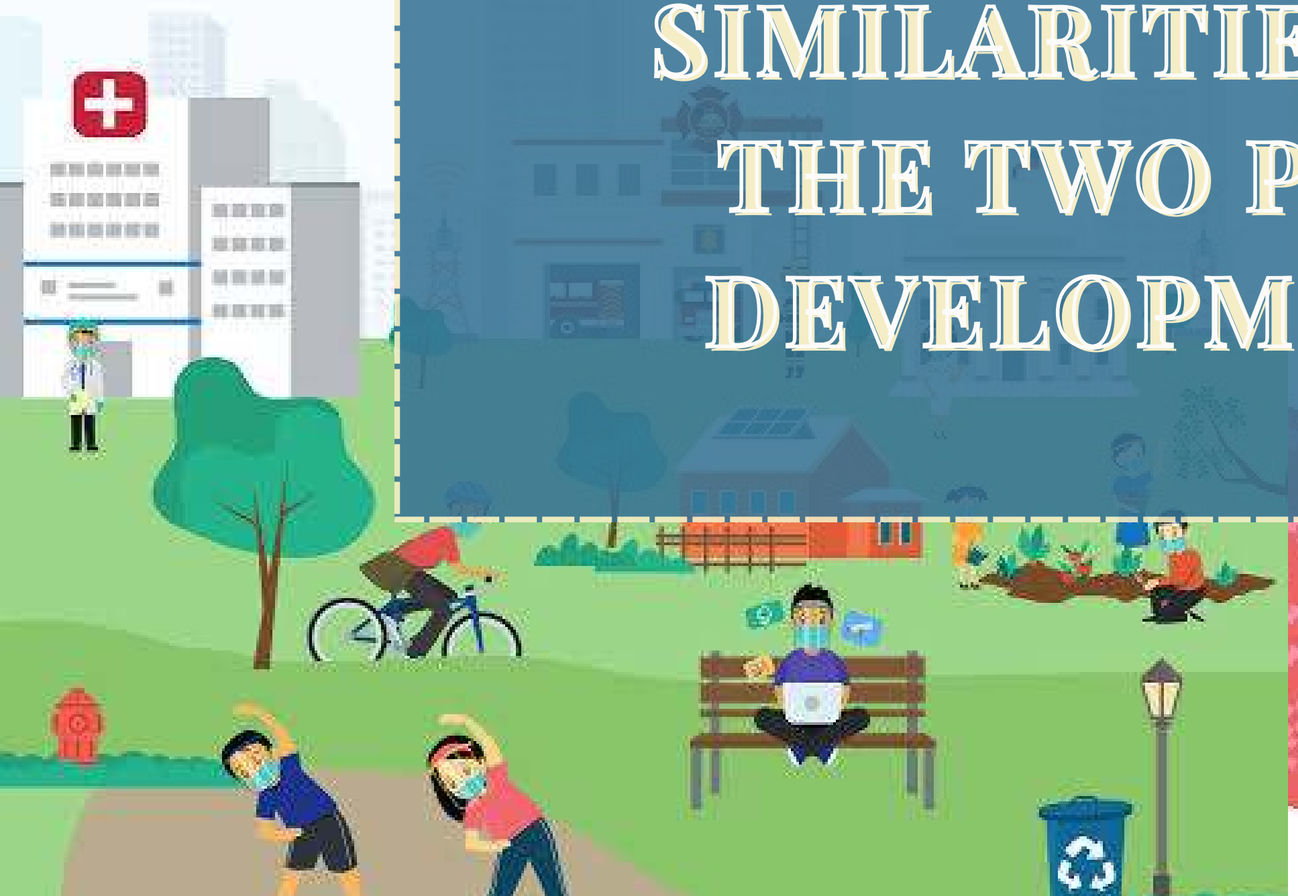
3 INCREASE INCOME-EARNING ABILITY

This strategy focuses on improving people's skills and qualifications through training and education to increase their employability. It also aims to expand employment opportunities by helping job seekers find suitable jobs and supporting workers in different industries.

UPDATED
Philippine
Development Plan
2017-2022

PHILIPPINE
DEVELOPMENT PLAN
2023 - 2028

SIMILARITIES BETWEEN THE TWO PHILIPPINE DEVELOPMENT PLANS



1. What is the common purpose of PDP 2017–2022 and PDP 2023–2028?

- Improve the Philippine economy and Filipinos' quality of life.
- Reduce poverty and create more economic opportunities.
- Promote inclusive growth that benefits all Filipinos

2. What do both plans aim to achieve for Filipinos in the long term?

- Support AmBisyon Natin 2040: "Matatag, Maginhawa, at Panatag na Buhay."
- Build a prosperous society with comfortable, secure, and healthy living conditions.

3. How do both plans address the economic and social needs of Filipinos?

- Economic: Create jobs, support local businesses, and improve access to markets.
- Social: Improve access to education, healthcare, housing, and essential services.

4. What common strategies do both plans use to achieve national development?

- Promote research and development, modern technology, and improved production processes.
- Strengthen business competitiveness and productivity through innovation.

UPDATED
Philippine
Development Plan
2017-2022

PHILIPPINE
DEVELOPMENT PLAN
2023 - 2028

DIFFERENCES BETWEEN THE TWO PHILIPPINE DEVELOPMENT PLANS



1. What are the differences in the main areas of focus between PDP 2017–2022 and PDP 2023–2028?

- PDP 2017–2022 focused on building a basic foundation for growth through three core pillars such as, enhancing social fabric, reducing inequality and increasing growth potential.
- PDP 2023–2028 focused on deep economic and social transformation aimed at rapid job creation and pushing national poverty down

2. What are the differences in the strategies used by both plans to promote national development?

- PDP 2017–2022: focused on infrastructure development, improved public services, and stronger institutions.
- PDP 2023–2028: focuses on modernizing industries, strengthening development, and promoting innovation.

3. What specific goals and targets make PDP 2023–2028 different from PDP 2017–2022?

- PDP 2017–2022: Targeted 7–8% economic growth, reducing poverty to 14% by 2022, and lowering unemployment to 3–5%.
- PDP 2023–2028: Targets 6.5–8% annual economic growth, reducing poverty to 8.8–9% by 2028, and improving the Philippines' global innovation ranking.

4. How does the financing and investment strategy for national projects differ between PDP 2017–2022 and PDP 2023–2028?

- PDP 2017–2022 is more focused on government spending for infrastructure projects.
- PDP 2023–2028 aims to attract investments from various sources.

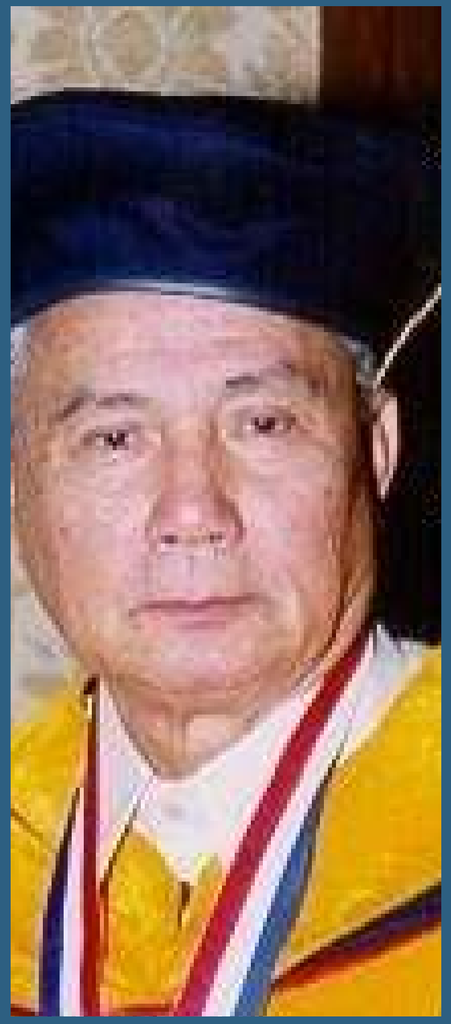
How the focus shifted from infrastructure and capacity building to digital transformation and innovation-led growth?

- PDP 2017–2022: Focused on developing infrastructure, improving public services, and strengthening the country's capacity for economic growth.
- PDP 2023–2028: Put greater emphasis on digital transformation and innovation to improve productivity, efficiency, and the competitiveness of Philippine industries.
- Builds on the previous plan by using modern technology and innovation to create more economic opportunities and support long-term economic growth.



NOTABLE FILIPINO SCIENTISTS









TEODULO M. TOPACIO

Specialization

Veterinary Medicine, Veterinary Microbiology, Animal Health, and Parasitology

Major Work/Invention

Pioneered research works on leptospirosis including its transmission from pigs to man. He also contributed his share in the research on animal diseases like brucellosis, fasciolosis and rabies and their control. He was also instrumental in the formulation of animal health policies and the enactment of the Philippine Anti-Rabies Act of 2007

Impact to Present Day

His study contributes to animal health, control of zoonoses, animal production, and rabies control programs in the Philippines.



MERCEDES B. CONCEPCION

Specialization

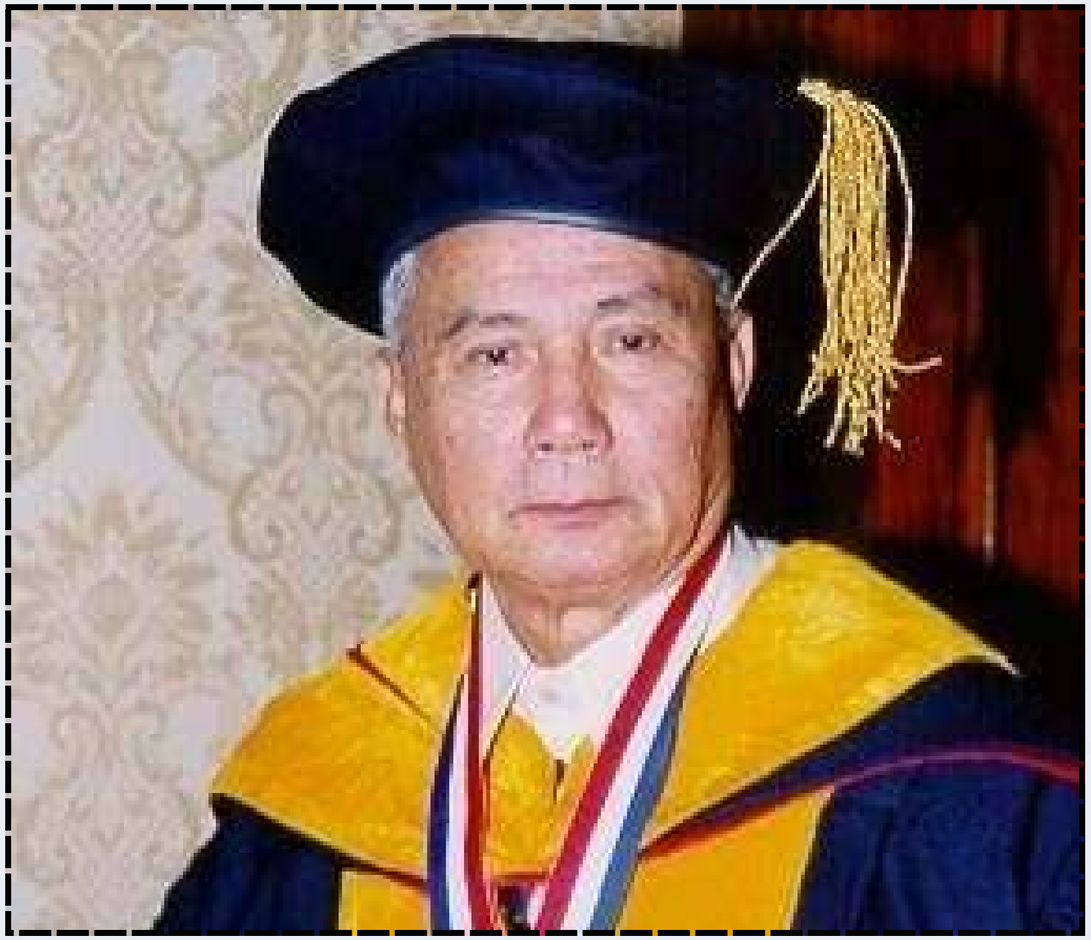
Demography, Sociology, and Statistics

Major Work/Invention

He was involved in research works on Hepatitis B and liver diseases in the Philippines. He also participated in studies and researches on hepatitis B control and prevention among others in the country.

Impact to Present Day

His study contributes to the control of hepatitis B and liver disease in the Philippines. It also promotes awareness, immunization programs, and research on hepatitis.



ANGEL C. ALCALA

Specialization

Biological Science and Biodiversity Conservation

Major Work/Invention

Pioneered community-based marine protected areas, with studies on the Philippines' marine ecosystem.

Impact to Present Day

His study aids in the protection of reefs, fisheries, and marine biodiversity which leads to sustainable coastal resources.



RAMON C. BARBA

Specialization

Horticulture and Plant Physiology

Major Work/Invention

He was the first to develop a practical method of inducing flowering of mango trees using potassium nitrate

Impact to Present Day

His study helps the production of mangoes, especially its flowering which leads to the reproduction of quality mangoes in the country.



GAVINO C. TRONO

Specialization

Marine Biology and Phycology (the study of algae)

Major Work/Invention

He did extensive research on seaweeds and helped develop seaweed farming in the Philippines.

Impact to Present Day

His study supports the seaweed industry, provides sustenance to coastal communities, and produces food and industrial products.

CONCLUSION

- **Early Filipino knowledge** — Traditional farming, medicine, and boatbuilding

- **Development of science and technology** — Education, research, and inventions

- **Contribution to society** — Filipino skills and knowledge in community and national development



CONCLUSION

IMPACT

- **Government programs** — Balik Scientist Program and K–12 STEM education

- **National development plans** — Technology, economic growth, and job opportunities

- **Filipino scientists' contributions** — Agriculture, healthcare, and environmental protection

- **Overall impact** — Practical solutions to community problems and national development



CONCLUSION

FUTURE OUTLOOK

- **Government support** — Better education, research opportunities, and funding

- **Technological solutions** — Addressing climate change and natural disasters

- **Youth and local researchers** — Encouraging future scientists and supporting local innovations

- **Combining traditional and modern knowledge** — Creating solutions for national challenges

- **Goal** — A more developed and resilient Philippines




THANK YOU!

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